ENGR 102 Summer Bridge
Day 14 Student Workbook

Lists, Length, Indexes, Mutation, and List Loops

Thursday, July 23, 2026 • 50 points

Name		UIN		Section	

Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Daily target — increasing independence

Choose index evidence when stored values must be read, changed, summarized, and located.

Allowed tools	Prior tools plus lists, len, append, indexed reads, and indexed assignment
Support supplied	A starting list and required final-state summaries are supplied.
You must decide	Loop form, list mutation, summary state, and first-largest tracking.
Scaffold level	Specification only; no planning prompts or variable inventory.

Scoring and completion standard — 50 points

Evidence	Points	Secure work shows
Integrated trace	10	intermediate state, condition results, and exact output
Diagnosis and repair	8	actual, intended, cause, repair, and a boundary test when requested
Full program and tests	32	coherent executable logic, required output, and defended tests

Independence standard. You may use the supplied requirements and examples, but you must produce one connected solution. A collection of disconnected correct lines is not yet a complete program.

Begin with the trace, then use its evidence habits when you construct.

Task 1. Integrated trace — 10 points

Trace index state, list values, an appended evidence list, and an accumulator in one ledger.

```
values = [6, 3, 9, 4]
total = 0
low_positions = []

for index in range(len(values)):
    if values[index] < 5:
        low_positions.append(index)
    else:
        total = total + values[index]

print(total, low_positions, values[len(values) - 1])
```

Your task

1. Give the complete index-by-index ledger and exact output, distinguishing each index from its stored value.

A final answer without the requested state evidence cannot earn full trace credit.

Task 2. Diagnose and repair — 8 points

The intent is to count every negative value, including the first and last positions.

```
values = [-2, 4, -1, 6, -3]
negative_count = 0

for index in range(1, len(values) - 1):
    if values[index] < 0:
        negative_count = negative_count + 1

print(negative_count)
```

Your task

1. Give actual versus intended result, identify both skipped boundaries, repair the loop, and name one compact test that protects both ends.

Debugging evidence is complete only when the repair is tied to the stated defect and an expected result.

Task 3. Construct a complete program — 32 points across Pages 4-5

Program specification

- Use readings = [12, -3, 7, 20, 7]. The list is nonempty.
- Loop by index and replace every negative value with 0.
- Using final values, compute valid_total and the number of values at or above 10.
- Find the index of the first largest final value without using max or index.
- Print the final list, valid_total, high_count, and largest_index on separate lines.

Program workspace – begin the connected solution here.

Task 3 continued. Test and defend the program **included in 32 points**

Continue code if needed.

Required tests, expected results, and brief defense

VIP Lab Alignment

Bridge Day 14 • Contract c821cfcf7189

This page adds a current lab handoff while preserving the workbook's independence-ramp tasks.

Why this page is here

VIP Lab Day(s): 6 **Activities:** 18.25, 18.26, 18.27

Carry index, value, and final-list evidence into reverse traversal and one- or two-condition filters.

Open these current notebook cells

Activity / stable cells	Notebook work	Evidence to carry forward
18.25 `18-25-work` / `18-25-transfer`	**Reverse Values**	Create a new list without mutating the input.
18.26 `18-26-work` / `18-26-transfer`	**Values At Or Above Cutoff**	Filter without reordering.
18.27 `18-27-work` / `18-27-transfer`	**Feasible Workstations**	Read supplied workstation records.

Readiness check before you code

1. State which mapped activity requires indexes and which activities can preserve order by traversing values. ____________

   ____

2. For Activity 18.27, write the two Boolean constraints separately before combining them with and. ____________

   ____

Notebook and submission procedure

1. Work in the provided sample and transfer cells; do not delete or recreate them.
2. Run each cell and compare fresh output with the notebook's stated result before revising.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.

Contract-controlled page • generated from the current VIP activity profiles and stable notebook cell IDs